Language Models

Brendan Tomoschuk

Updates to class schedule

Week 8

- **Monday:** [remote, recorded] **Interview with Dr. Jolene Mork from Microsoft AI**
- **Wednesday:** [in person] Andrew Cronan - Career Decision Making
- **Friday:** Class is CANCELLED

Week 9

- **Monday:** Midterm 2
- **Wednesday:** Class will be REMOTE OVER ZOOM
- **Friday**:No lecture, no section (Thanksgiving)

Midterm 2 — Monday November 23

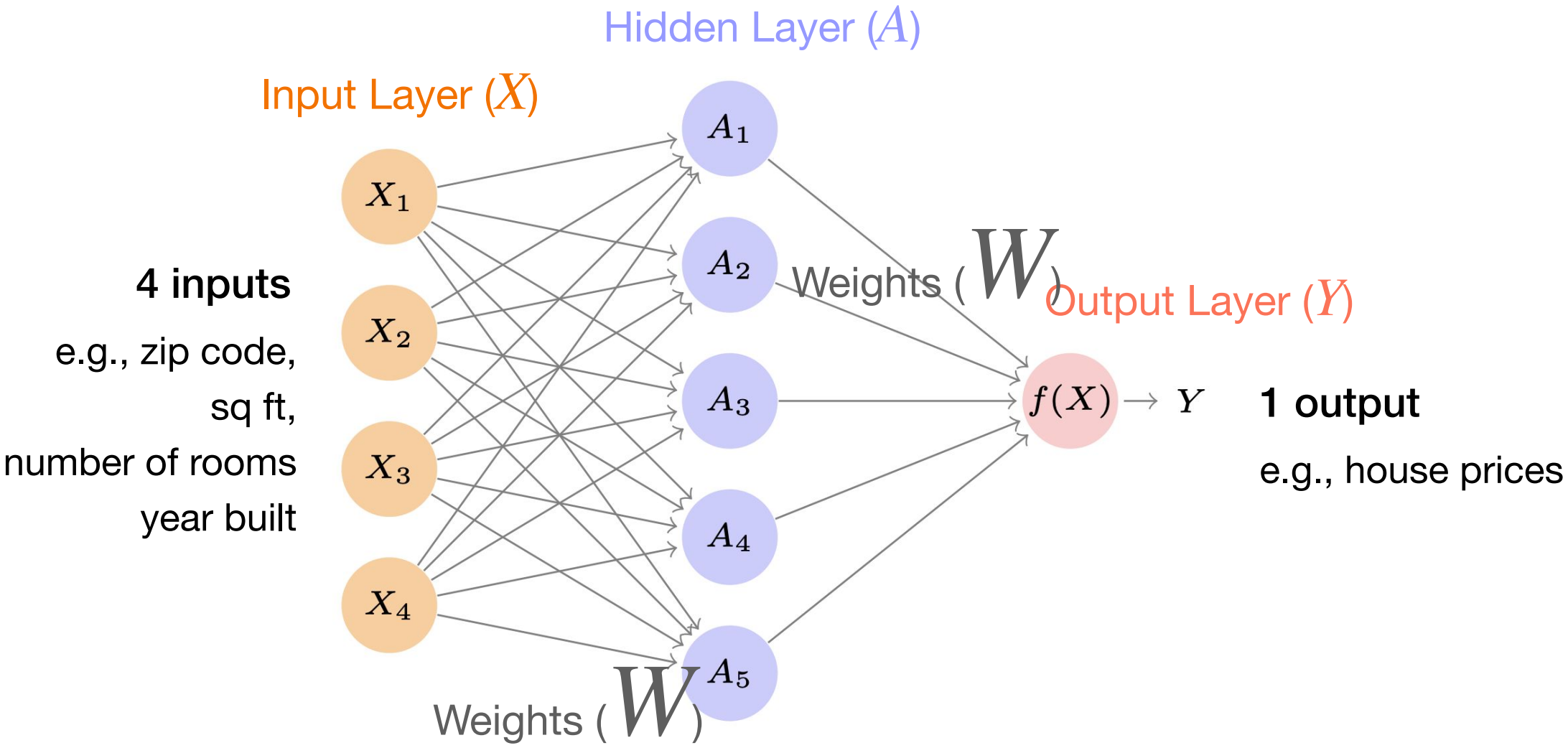
- **Format:** 35 multiple-choice questions, 50 mins, ***CLOSED-NOTE***
    - ***Questions taken from***
        - ***lectures from "Data visualization" to "Language Models"***
        - ***Readings 4-5***
- **Bring:**
    - A pen or pencil
    - No need to bring a scantron
- **Reminders:**
    - Come early and get a seat to maximize everyone's time!
    - Students with accommodations will take the exam at the Triton Testing Center.

Learning Goals

1. Define two language models and give examples
2. Describe the input and output of a language model **(tokens → next-token probabilities)**
3. Explain how language models are trained **(next-token prediction on lots of text)**
4. Describe their capabilities, limitations (hallucination, bias) and evaluation

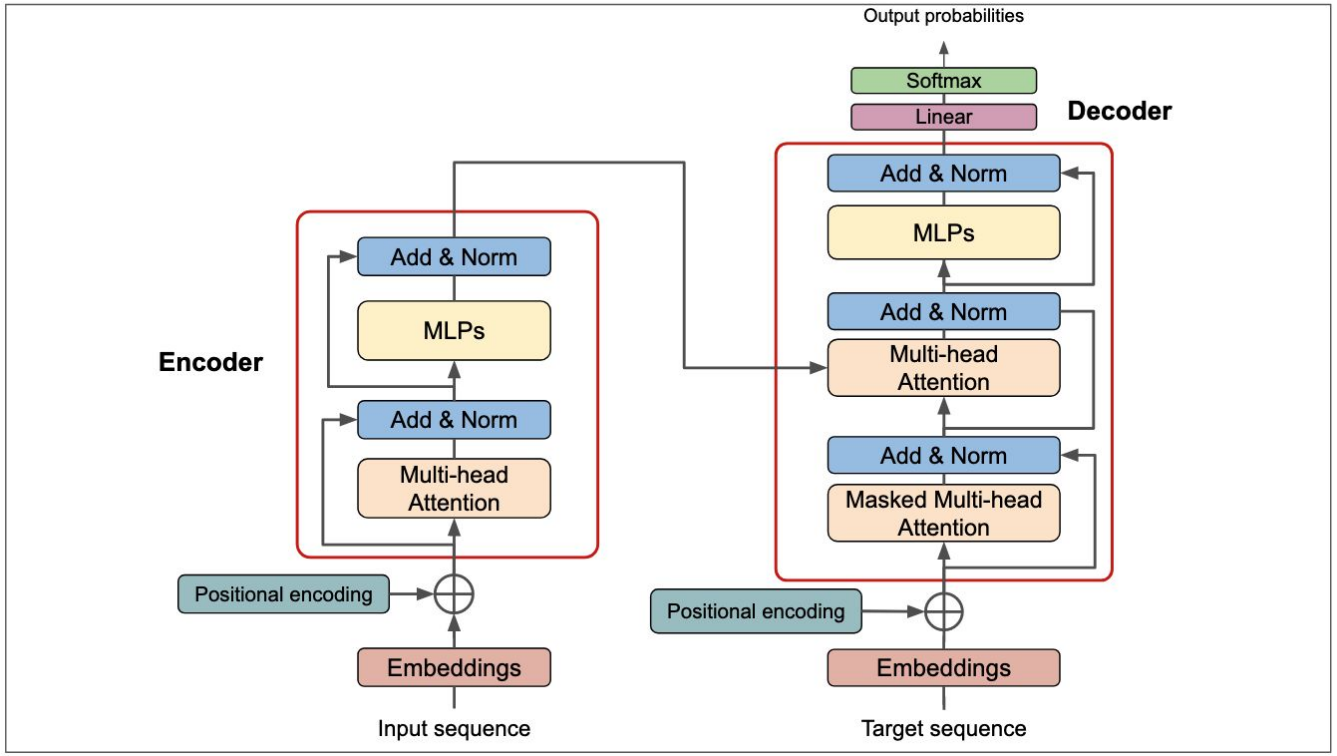
Neural Networks

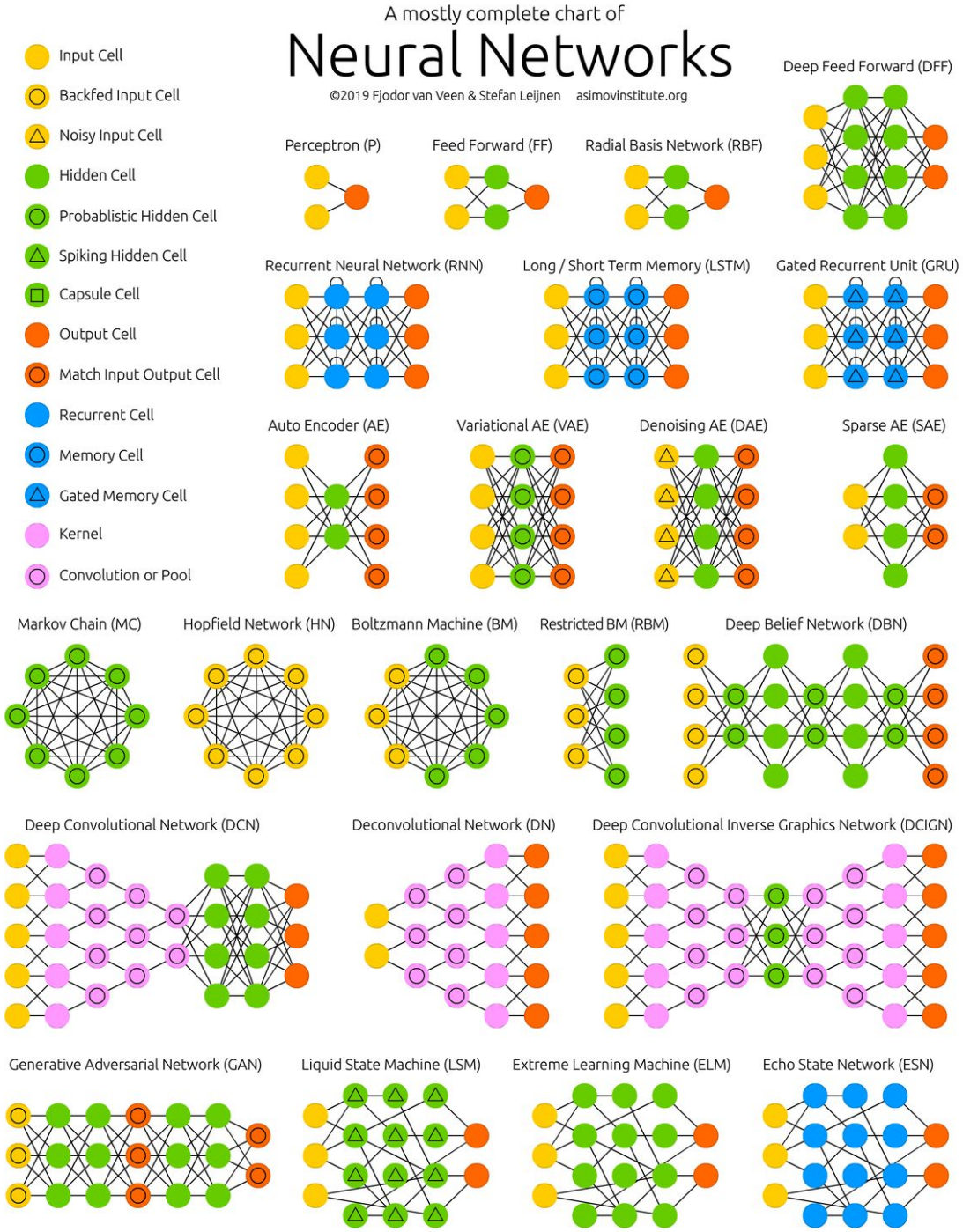
Goal: estimate the weights (W) that relate the inputs (X) and outputs (Y)



There are many, many types of neural networks…

Large language models (LLMs) like ChatGPT are a special type of neural network called **transformers:**





Today: Language Models

Modeling language in order to **predict what comes next** and **generate fluent text**.

What is a language model?

A language model learns the patterns of language from huge amounts of text, called a **corpus.**

A corpus for a Large Language Model might require BILLIONS of words and cost MILLIONS of dollars to train

The Times Sues OpenAI and Microsoft Over A.I. Use of Copyrighted Work

Millions of articles from The New York Times were used to train chatbots that now compete with it, the lawsuit said.

Share full article 1.3K



KATE KNIBBS BUSINESS OCT 7, 2024 7:30 AM

The Race to Block OpenAI's Scraping Bots Is Slowing Down

OpenAI's spree of licensing agreements is paying off already—at least in terms of getting publishers to lower their guard.



Input: text → tokens

Input is plain text — but the model doesn't see words directly

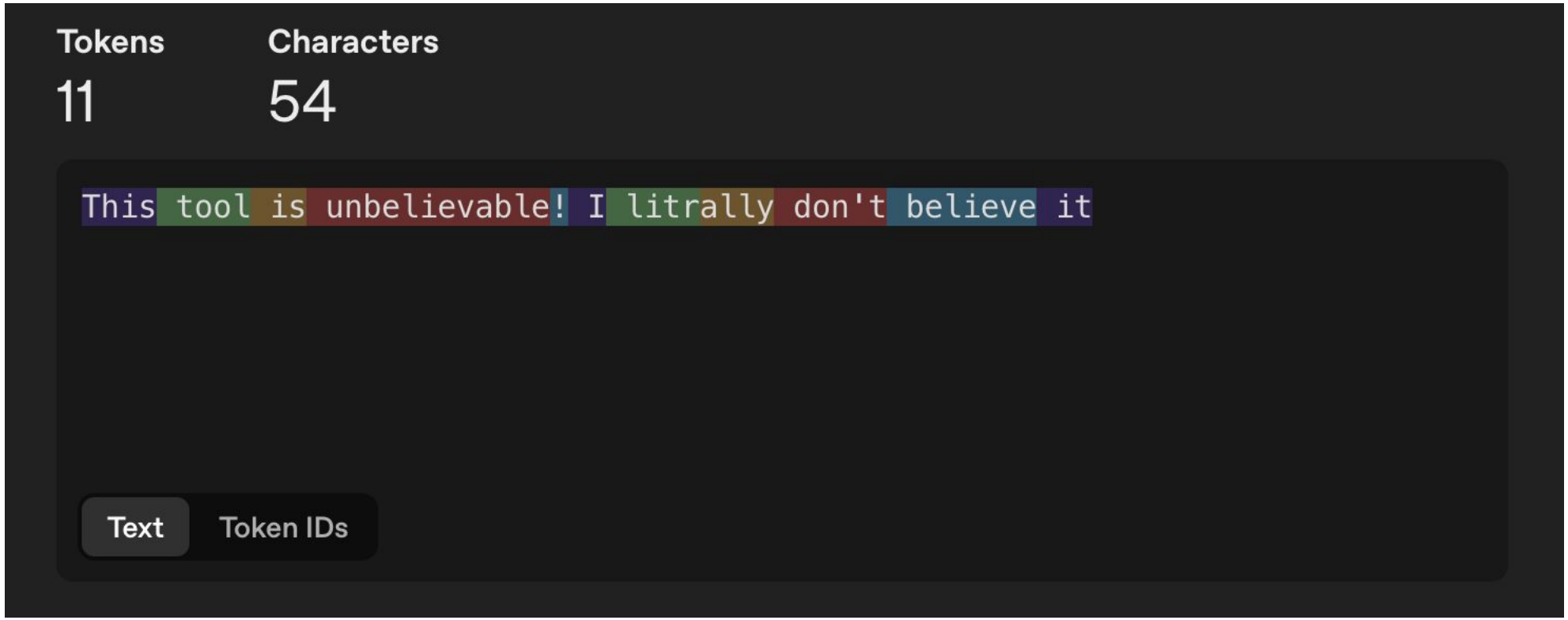
The text is broken into **tokens** — small units (a word or part of a word)

Example: "The quick brown fox jumped over the lazy dog" → **[The] [quick] [brown] [fox]**

Tokens: Often parts of words

Subword tokens let the model handle **new, rare, or misspelled words** and other languages

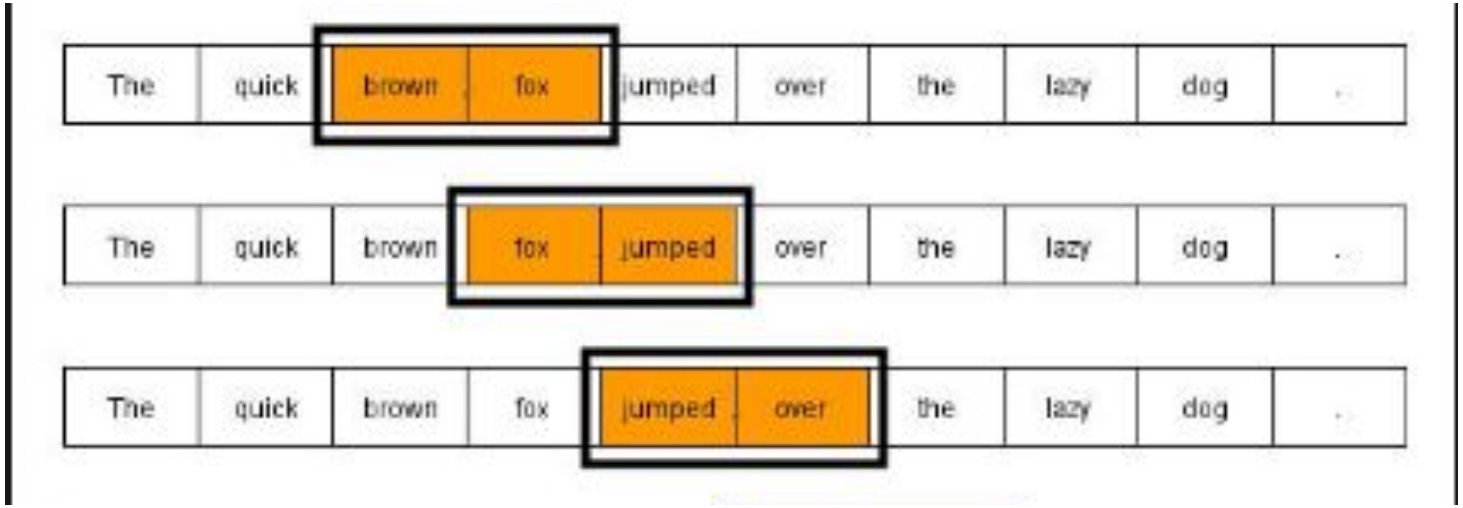
Example: OpenAI Tokenizer https://platform.openai.com/tokenizer



(Small) Language Models: N-grams

An **N-gram** model looks N words back to predict the next word

- [N = 1] Given all of the text in the corpus what is the probability that the word fox comes after "**brown**"?
- [N = 2] Given all of the text in the corpus what is the probability that the word fox comes after "**quick brown**"



Output: next-token probabilities

The model outputs a **probability** for **every token in the vocabulary**

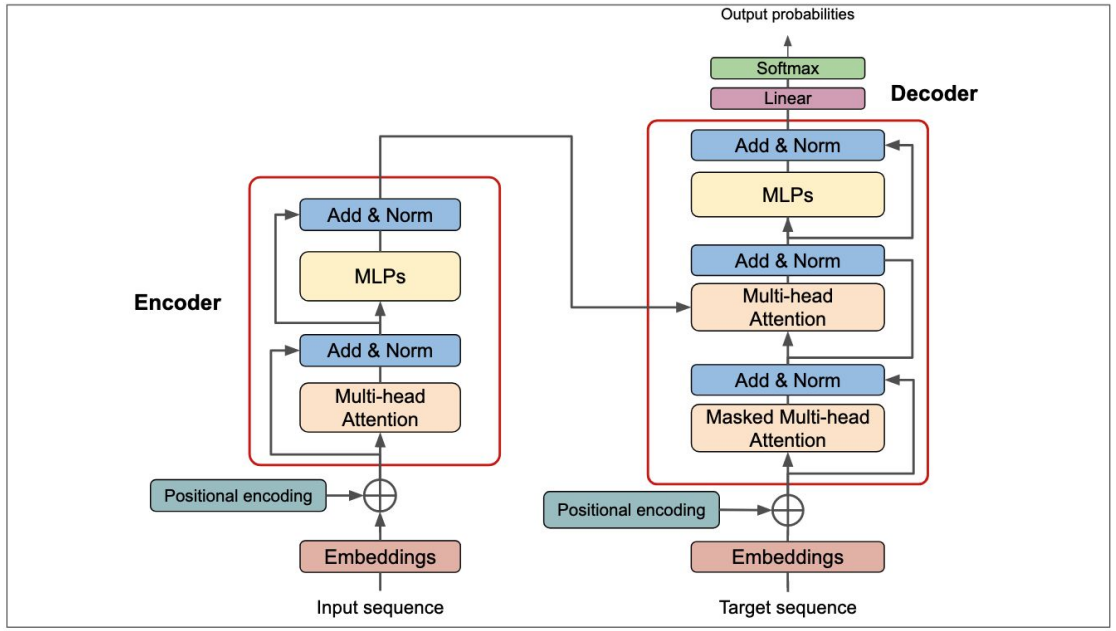
Likely next words get **high** probability

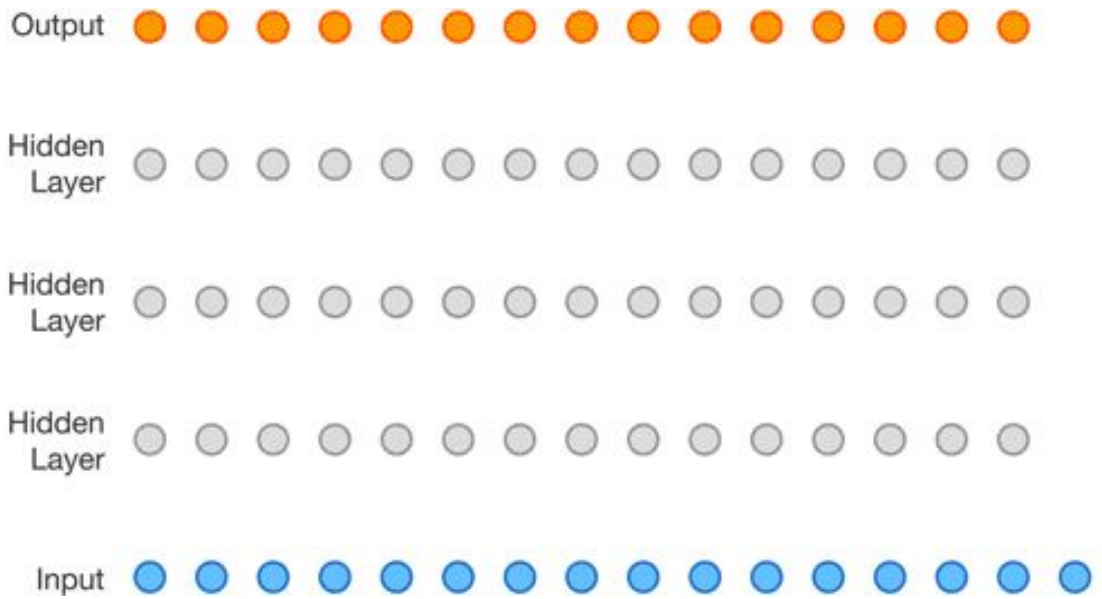
The quick brown fox ___

jumps 0.42	runs 0.31	eats 0.18	sleeps 0.07

Large Language Models (LLM)

A Large Language model takes the idea behind an N-gram model and uses a Transformer and makes it self-supervised.





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course code:

WSTAPX

Which of the following best describes what a language model is doing?

(A) Looking up words from a table

(B) Looking up tokens from a table

(C) Predicting what word will come next

(D) Predicting what token will come next

(E) I'm not sure / I'm confused

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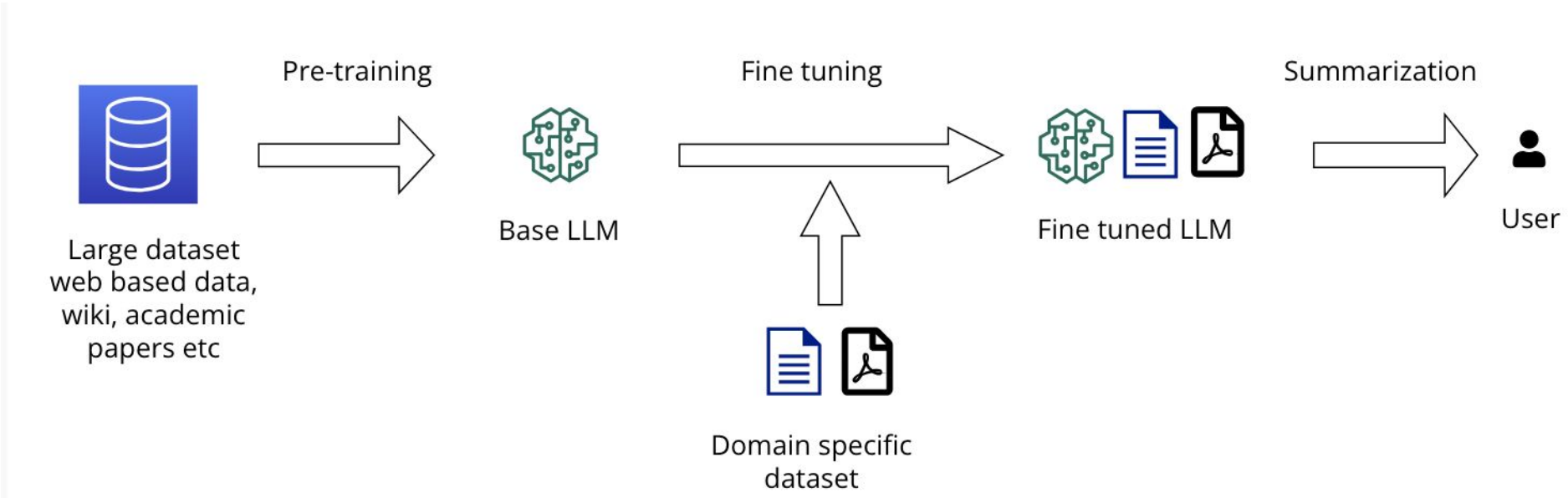
A useful LLM needs to do more

1. **Pretrain** [what we've discussed so far] — next-token prediction on big body of text
2. **Fine tuning** — Apply a **supervised** learning task to the pre-trained model
3. **Post-training & Alignment** — humans rank responses to try to make the model more **helpful, harmless, honest**

Fine tuning

Examples:

- **Legal case data (X) and summaries (Y)**
- **Customer service requests (X) and responses (Y)**



Post-training & Alignment

How do I fix a leaky sink?

After pre-training

- Call a plumber. - Check under the sink for tools. - How to install a new faucet in 5 easy steps...

After fine-tuning

To fix a leaking sink, first turn off the water supply under the basin. Next, place a bucket underneath the P-trap and unscrew the slip nuts to inspect for debris or a worn rubber washer."

After post-training

Here are the steps to fix a leaking sink safely:

Shut off the water. valves beneath the sink.

Check the P-trap connections for loose nuts.

Safety note: If water is spraying near electrical outlets, turn off your main home water supply immediately.

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course code:
WSTAPX

A pre-trained LLM is _____, while a fine-tuned LLM is _____

(A) self-supervised; supervised

(B) unsupervised; self-supervised

(C) unsupervised; supervised

(D) supervised; self-supervised

(E) I'm not sure / I'm confused

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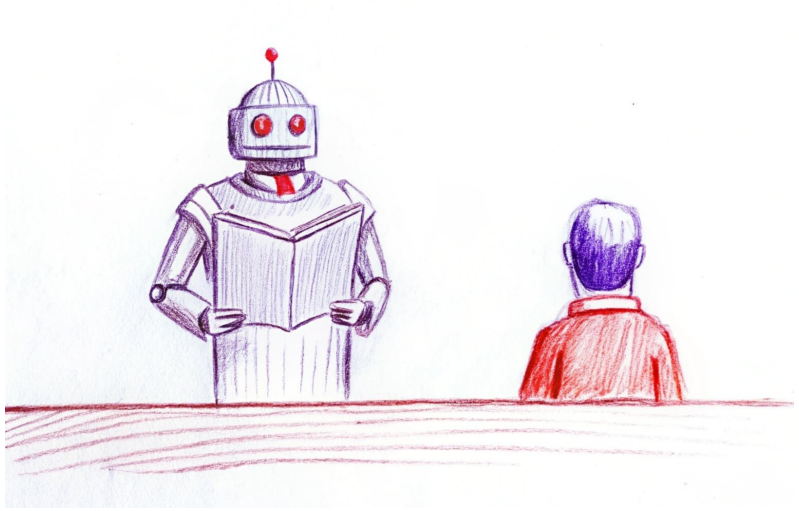
(D) supervised; self-supervised

(E) I'm not sure / I'm confused

Limitations: hallucination

Models predict **plausible** text, not **verified facts**

They can be confident and wrong — this is called **hallucination**



"During the New York federal court filing, one of the lawyers was caught citing non-existent cases. It turned out he was using ChatGPT to conduct legal research — the bot referenced **fake cases** to the attorney."

Limitations: Bias

Trained on human text → can reflect the bias already in that text

Examples:

- Stereotypical Occupation Associations: Mislinking pronouns based on gender stereotypes in professions, e.g., assuming doctors are male

- Regional Language Varieties and Dialects: Struggling with comprehension of non-standard dialects

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course code:
WSTAPX

A chatbot confidently reports that a fact that further research shows is completely made up. This is referred to as:

(A) overfitting

(B) bias

(C) hallucination

(D) undherfitting

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course code:

WSTAPX

Why can a language model produce biased or offensive text?

(A) It was trained on human text that contains bias

(B) It is programmed with explicit bias rules

(C) It only works for a single language

(D) It stores personal data about users

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Discuss with a neighbor (~5 min)

LLMs seem to “know” facts, but we just discussed that at its core it’s predicting the next token. Do you think LLMs can know things? How might we test whether an LLM has knowledge rather than just good predictions?

Enter cognitive science!

Example 1: Turing Test

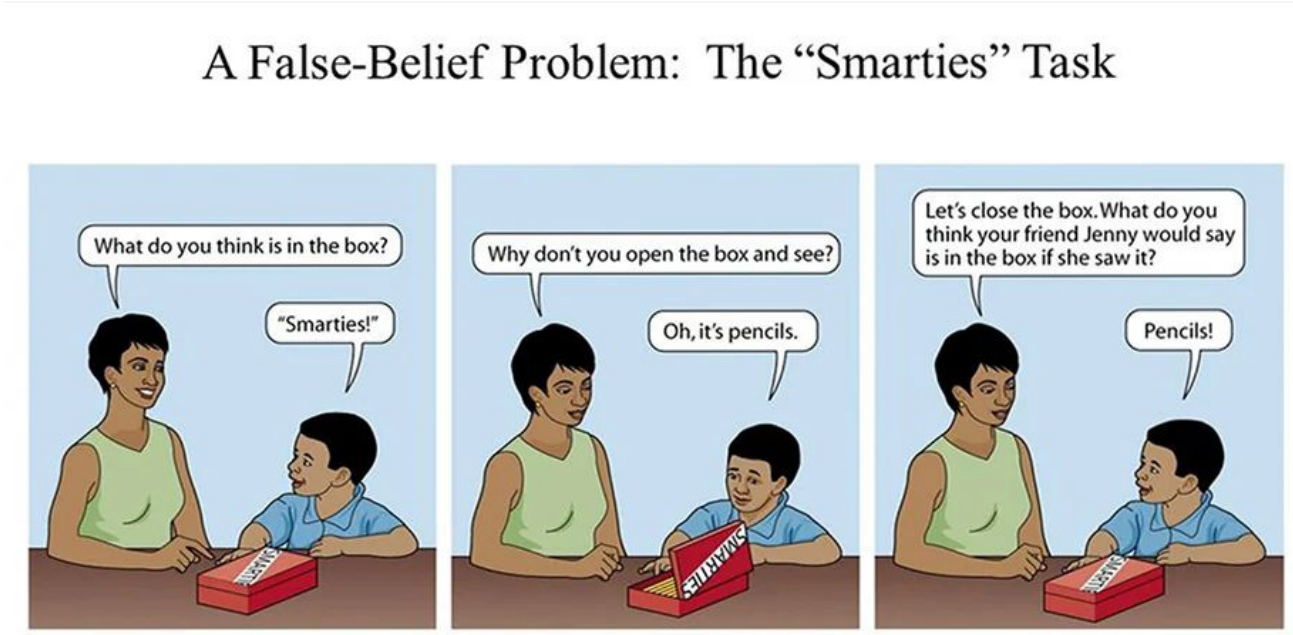
Researchers at UCSD found that when prompted to adopt a humanlike persona, **GPT-4.5 was judged to be the human 73% of the time**: significantly more often than interrogators selected the real human participant. -Jones & Bergen 2026

COGS 150: Large Language Models & Cognitive science



Enter cognitive science!

Example 2: Theory of Mind



"Results are mixed. LLMs display considerable sensitivity to mental states and match human performance in several tasks" - Jones, Trott & Bergen 2024

LLM evaluation

These kinds of tasks (along with many others from both cognitive and computer science) are used as benchmarks for LLM evaluation.

Building evaluation tasks is almost as hard as building the model itself!

